

Project Proposal: Matsuri Nights — A Japanese Festival Street

Course: Computer Graphics and Image Processing Laboratory (CSE-4102)

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Project Overview

This project presents a real-time interactive 3D recreation of a traditional Japanese summer street festival (matsuri), rendered in both a daylight and a nightlight lighting pass. A single festival street, lined with Japanese-style wooden machiya buildings and ending at a torii gate, is strung with swaying paper lanterns and lined with food stalls, a small magic-show area, and a walking crowd, with fireworks lighting up the night sky. The scene is designed to demonstrate the full set of 3D model transformations, hierarchical (parent-child) transformations, and Phong-based illumination and shading covered in the course, while remaining a single compact street scene. Multiple independently moving and transforming objects and light sources are used specifically to show how transforming an object relative to another object's reference frame, and moving/transforming a light source, visibly change the scene's shading and color.

Scene Objects

1. Machiya Buildings and Torii Gate (static reference architecture): Two-story Japanese wooden building facades with sloped tiled roofs and sliding shoji windows line both sides of the street, ending in a large red torii gate that anchors the scene. Built from combinations of boxes and pitched-roof prisms, these objects remain static and serve as the fixed reference frame against which every moving object in the scene is judged; their shoji windows switch from dark (day) to a warm emissive glow (night).

2. Paper Lanterns (hierarchical, complex moving light source): Rows of red chōchin lanterns are strung on ropes across the street. Each lantern is a child node of a short rope node, which is itself a child of the building/cart it hangs from, and the rope's local rotation oscillates to produce a pendulum swing. This is the project's core relative-transform demonstration: the lantern's world position is computed entirely through this parent-child chain. Unlit by day, each swinging lantern also acts as a warm point light at night, so its motion directly changes the lighting on nearby surfaces.

3. Food Stalls — Takoyaki and Kakigori (detailed, moving parts): Two yatai stalls stand along the street. The takoyaki stall's grill holds several small balls that rotate in place and arc-translate between molds to simulate flipping; the kakigori stall has a rippling cloth noren curtain (vertex-based sine-wave animation) and a fluttering nobori flag. Each stall has its own local lantern/light, giving a lighting/color contrast point between the two stalls.

4. Magician and Two Magic Tricks (complex hierarchical + particle-style moving objects): A magician figure, built as a hierarchical rig (torso → upper arm → forearm), performs two tricks on a small stage. In the Floating Orb trick, a glowing orb orbits the magician's raised hand along a helical path computed relative to the hand's own moving transform — the strongest example of transforming an object relative to another object's reference frame — while also acting as a small moving point light. In the Vanishing Box trick, a box scales to zero beneath a rippling silk cloth and reappears at a

second stage position, demonstrating a scale-to-zero transform combined with a discrete position swap. A spotlight above the stage tracks the magician, so its transform visibly changes the shading on him and the orb.

5. Crowd Figures (independent moving objects): Several simple yukata-wearing figures walk along the street between the stalls and torii gate, each following its own translated path and looping back at the street's end, adding independent, continuous motion throughout the scene.

6. Fireworks (night sky, particle-style complex motion): Each firework launches upward as a single glowing shell before exploding into a burst of small colored particles that radiate outward under gravity and fade over time. Several fireworks are staggered with different colors and timings, and each explosion briefly acts as a short-lived colored point light, tinting nearby lanterns and building facades — a clear demonstration of a moving/changing light source altering surrounding object colors.

Illumination and Shading

The scene uses the Phong illumination model (ambient + diffuse + specular components), computed per-fragment for Phong shading across all surfaces. A directional sun/moon light sweeps and blends in color, angle, and intensity to drive a full day-to-night transition, while multiple point and spot lights — the swinging lanterns, stall lights, the tracking stage spotlight, the orbiting orb, and the momentary firework bursts — provide localized, moving illumination. Because several of these lights are directly attached to moving/transforming objects, the demonstration clearly shows how repositioning or transforming a light source changes the color and shading of nearby geometry in real time.

Extra Features

- Interactive camera (keyboard/mouse) to explore the street and inspect transformations and lighting from any angle.
- Day/night toggle to trigger the full lighting transition on demand during the demo.
- Pause/resume control for all scene animation, to freeze and inspect any transform mid-motion.
- Falling sakura (cherry blossom) petals near one building corner for added ambient motion.

Interactive Demonstration Capabilities

To accommodate live, in-class testing, every object and light in the scene is built on a single, centralized Transform component (position, rotation, scale) within the scene graph described above. This means any object can be manually translated, rotated, or scaled relative to another object's reference frame on request during the demo, any light source can be repositioned or re-transformed live to show its effect on nearby colors and shading, and the day/night state, lantern lights, and firework bursts can all be triggered on demand — directly supporting the kind of live "transform this object," "change the lighting," and "move the light source" requests the instructor may ask for during evaluation.

Development Environment and Tools

The project will be implemented in C++ using modern OpenGL for rendering. GLFW 3.5.1 is used for window creation, input handling, and context management, while GLAD handles OpenGL function loading and GLM provides matrix/vector mathematics. Visual Studio serves as the primary IDE for development, debugging, and testing. Development follows three phases: Phase 1 builds all

geometry (five base primitive meshes composed into every object), the scene graph, and all animation logic using flat per-object colors; Phase 2 adds the Phong illumination pipeline with directional, point, and spot lights and the day-night state machine; Phase 3 applies UV-mapped textures (wood, roof tiles, lantern paper, cloth patterns) to finalize the visual aesthetic without disturbing the transformation or lighting logic.

Summary

Overall, the project remains a single, compact festival-street scene while covering every required concept: hierarchical/relative transformation (lanterns on their ropes, the orb orbiting the magician's hand), independent and particle-style complex motion (crowd figures, stall details, fireworks), Phong illumination/shading, and multiple moving light sources whose transformation visibly changes the scene's lighting and color, alongside a complete day-to-night lighting transition.